Run report: teske_variants_field_types, run 20261018T0845Z

Descriptive record of one run of `examples/teske_variants_field_types.rs` (protocol `research/teske_variants_field_types_20261018/PROTOCOL.md`, frozen at commit `62b4502c7`). It lists what ran, on what host, and the values the tool recorded, with their files. Interpretation is in `REPORT.md` of the study. Nothing here is a running time of an attack; the only timings are the tool's own wall-clock seconds per part, from `tool_log.txt`.

Command, host and isolation

```
run: 20261018T0845Z (Part C only)
protocol commit: 62b4502c7
why a second run: run-20261018T0630Z completed Parts A and B and Part C for q <= 11, then the session's worker
process was restarted and the tool was killed during q = 13 (see its NOTE.md). Before this run the census root search
over F_{q^6} (a scan of the field) was replaced by Cantor-Zassenhaus splitting with polynomial arithmetic over the
same table field; the census files for q <= 11 are byte-identical between the two tools (verified with cmp before
this run), and this run supplies Part C for all q.
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.80GHz, 15 GB
command: rustc --edition 2021 -O examples/teske_variants_field_types.rs -o tvft && ./tvft research/
teske_variants_field_types_20261018/results/run-20261018T0845Z C
tool sha256: 98ff1e88eea94c083d694abae25375f3df924280ea860d9b655392c96f705a10 (rustfmt-formatted; the committed file)
threads: 4 (std::thread::scope)
```

Isolation: none earned (a shared cloud container, 4 threads, no pinning); the tool is exact, so isolation affects only the wall-clock figures below.

Tool log (wall-clock per part)

```
part C q=3 models=5832 res_shaped=152 classes=21 sig=6 nosig=15 na=0 enum=0.0s total=0.0s
part C q=5 models=187500 res_shaped=3088 classes=158 sig=20 nosig=138 na=0 enum=0.1s total=0.2s
part C q=7 models=3294172 res_shaped=48352 classes=538 sig=39 nosig=499 na=0 enum=1.8s total=3.1s
part C q=11 models=42517464 res_shaped=293440 classes=3173 sig=106 nosig=3067 na=0 enum=49.0s total=62.6s
part C q=13 models=135150652 res_shaped=682472 classes=5582 sig=160 nosig=5422 na=0 enum=195.1s total=231.5s
elapsed 297.6s
```

Part A: not part of this run

Part B: not part of this run

Part C: census (`census.csv`, `census_summary.csv`, `census_patterns.csv`, `census_checks.txt`)

q	models	square-free	Res-shaped models	classes	signa-ture	no sig-nature	roots outside F_q^6	#E prime	prime, no sig.	t sub-field-shaped	of which sig.	traces	traces with sig.	s
3	5832	3888	152	21	6	15	0	0	0	7	0	9	2	0.0
5	187500	150000	3088	158	20	138	0	0	0	70	8	21	10	0.2
7	3294172	2823576	48352	538	39	499	0	0	0	92	1	37	9	3.1
11	42517464	38652240	293440	3173	106	3067	0	0	0	547	6	73	16	62.6
13	135150652	124754448	682472	5582	160	5422	0	0	0	1042	28	93	36	231.5

Classes by Weierstrass cycle type and signature:

q	cycle type	signature	classes
3	1+1+3+3	no	2
3	1+1+3+3	yes	4
3	1+1+6	no	11
3	2+6	no	2
3	2+6	yes	2
5	1+1+3+3	no	30
5	1+1+3+3	yes	12
5	1+1+6	no	76
5	2+3+3	no	12
5	2+6	no	20
5	2+6	yes	8
7	1+1+3+3	no	116
7	1+1+3+3	yes	19
7	1+1+6	no	211
7	2+3+3	no	38
7	2+6	no	134
7	2+6	yes	20
11	1+1+3+3	no	832
11	1+1+3+3	yes	54

11	1+1+6	no	1168
11	2+3+3	no	264
11	2+6	no	803
11	2+6	yes	52
13	1+1+3+3	no	1264
13	1+1+3+3	yes	92
13	1+1+6	no	2198
13	2+3+3	no	528
13	2+6	no	1432
13	2+6	yes	68

Checks (`census_checks.txt`):

```
q=3 models=5832 squarefree=3888 res_shaped_models=152 classes=21 neighbours_missing=0
q=3 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=5 models=187500 squarefree=150000 res_shaped_models=3088 classes=158 neighbours_missing=0
q=5 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=7 models=3294172 squarefree=2823576 res_shaped_models=48352 classes=538 neighbours_missing=0
q=7 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=11 models=42517464 squarefree=38652240 res_shaped_models=293440 classes=3173 neighbours_missing=0
q=11 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
q=13 models=135150652 squarefree=124754448 res_shaped_models=682472 classes=5582 neighbours_missing=0
q=13 check8_hasse_violations=0 t_inconsistent_in_class=0 check9_signature_numE_mod4_violations=0
check9_signature_cycle_violations=0
```

Graphs

Recorded values only, drawn by `figures/plot.py` from the CSVs of the two recorded runs (Parts A and B from run 20261018T0630Z, Part C from run 20261018T0845Z).

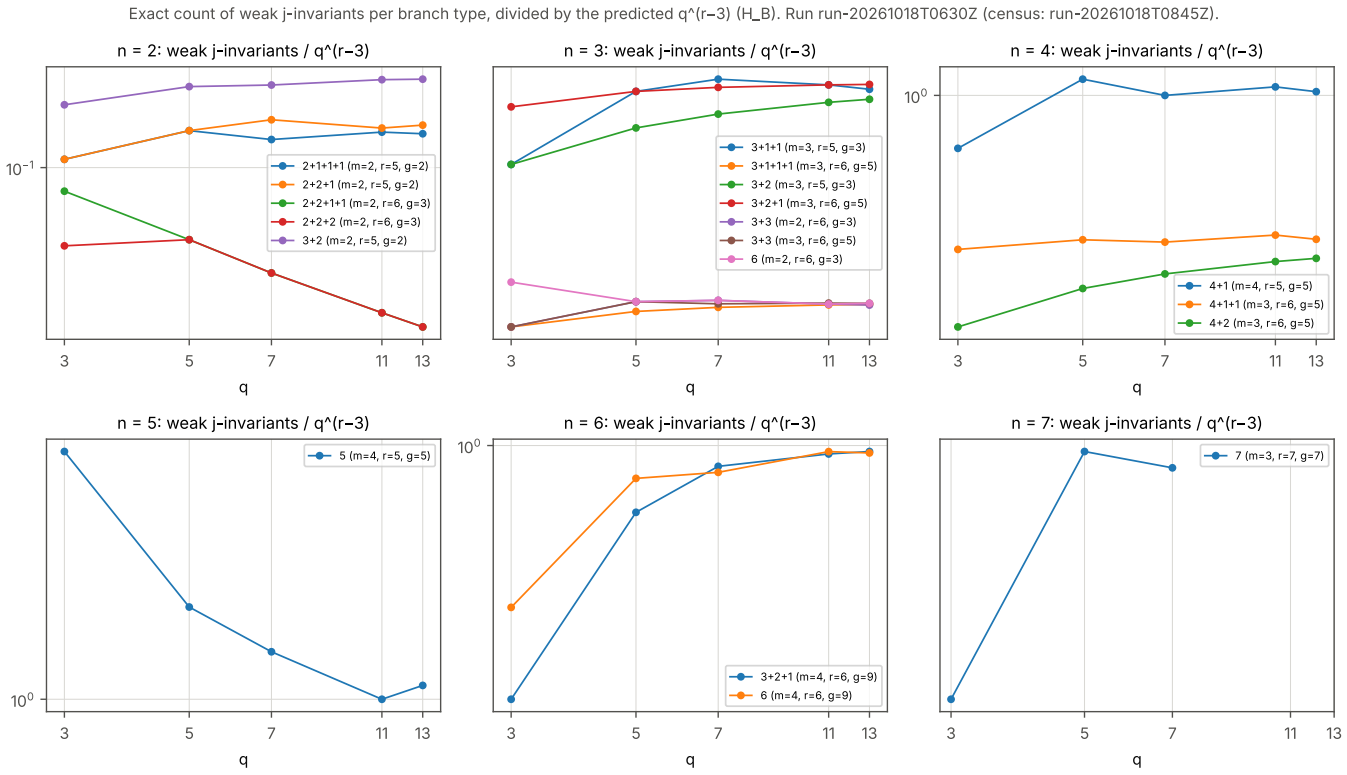


Figure 1: Weak j-invariants per branch type divided by q^(r−3); exact counts, no uncertainty.

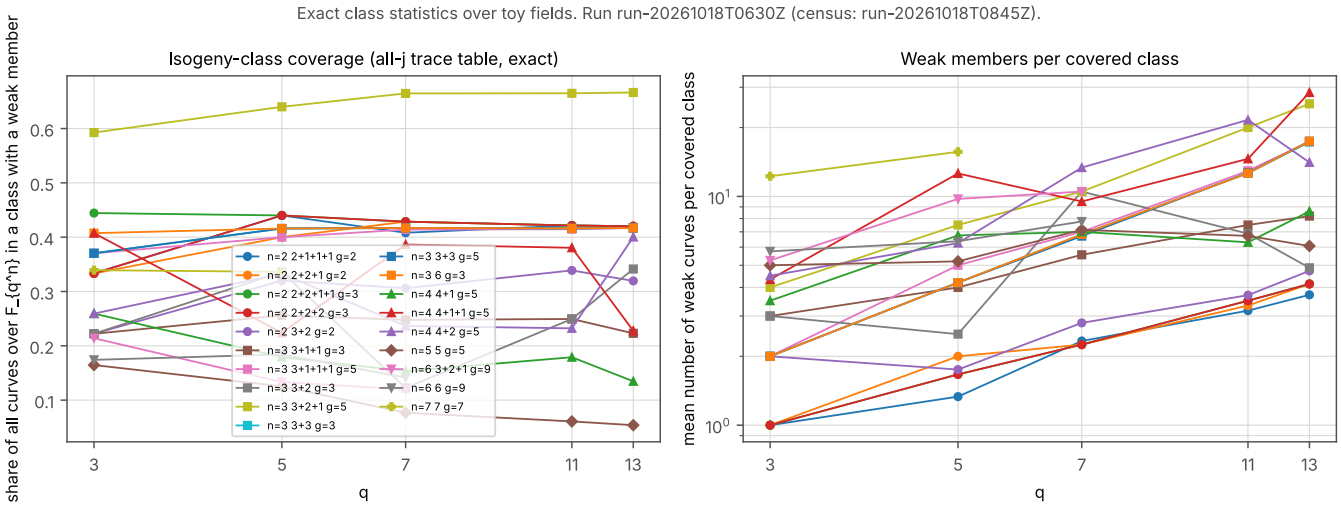


Figure 2: Share of curves in isogeny classes with a weak member, and weak members per covered class; exact, from the all-j trace tables.

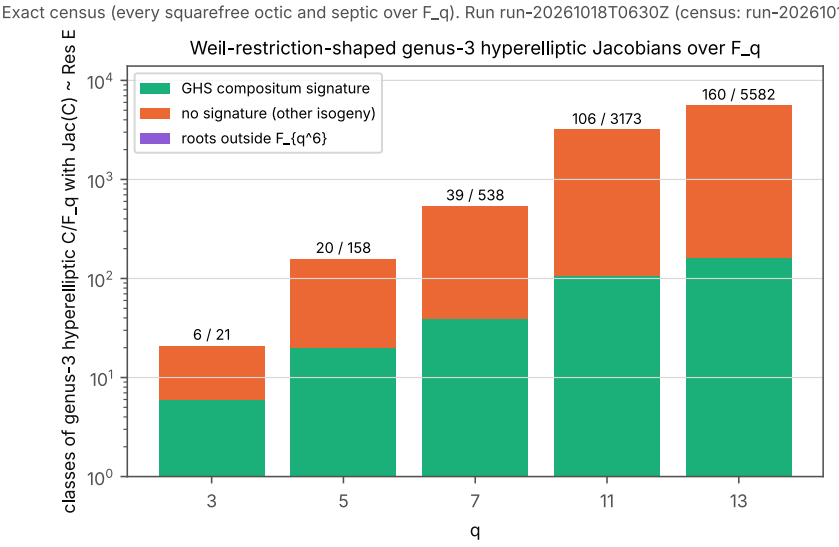


Figure 3: Res-shaped genus-3 hyperelliptic classes over F_q by GHS signature; exact census.

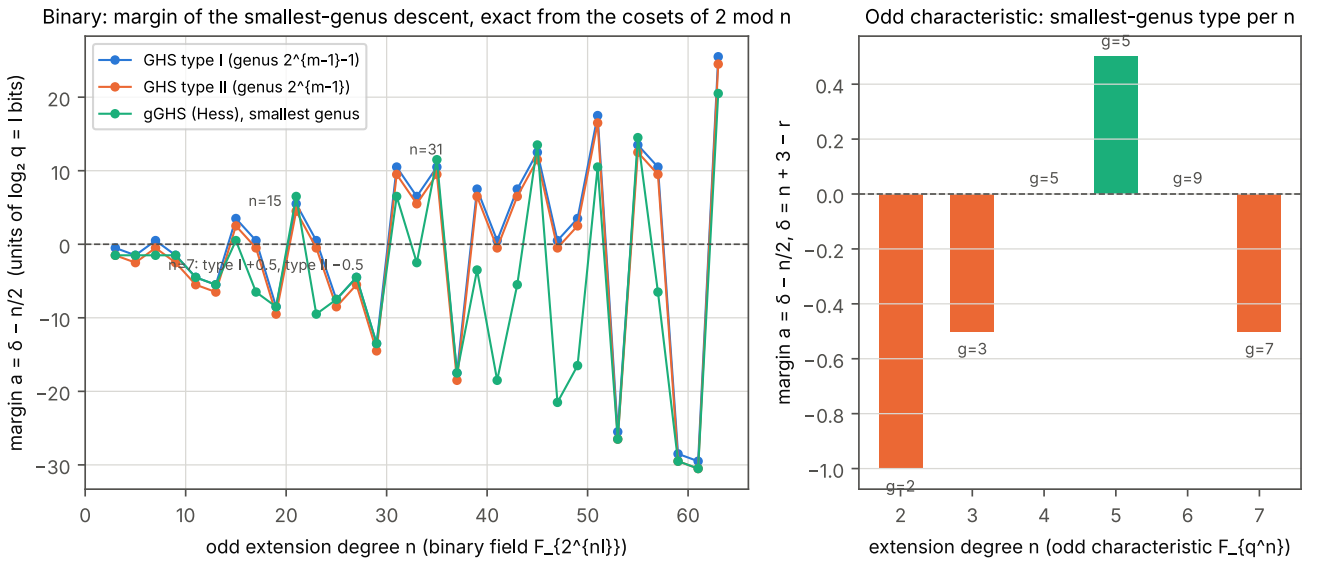


Figure 4: Margins a = δ − n/2 by n: binary (exact cosets) and odd characteristic (smallest-genus type).